

Family 1.gf: the global fine-granularity observation family
a dataset design note — what is measured at ten kilometres and
five days or finer, and how it is stored
raw observations first; derived products only where they carry raw observations nothing
else does

blauewelt/earth — design note

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Abstract

Family 1.gf is a proposed family of observation stores covering the whole globe at a granularity of **ten kilometres or finer in space and five days or finer in time**, built on the storage contract of family 10 (one registry, three tiers, one token carrying value, offset and footprint). It extends family 10's five point stores — 10.1's drifters, tropical moorings, ship CO₂ and 2.03 billion along-track sea-level samples, and 10.2's fishing-effort store (E-081, the fleet as a distributed sensor) — with the fine-scale measurements the programme's four prediction goals (El Niño, the ocean carbon sink, the AMOC, and sea-surface temperature) can each be argued to need: daily 4 km ocean colour, daily 5 km and 2 km sea-surface temperature from the instruments rather than the analyses fitted to them, 6 km sea-ice concentration, 2 km swath sea level from SWOT, 4 km cloud-top temperature every 30 minutes, ship reports back to 1662, every non-Argo ocean profile since 1772, interior carbon bottles, biogeochemical floats, open-ocean moorings, and satellite column CO₂. Every candidate was checked against its producer on 2026-09-16: resolution, cadence, record, access, volume and licence. Three things the research settled change the design: no ocean-wind, sea-surface salinity or microwave-SST product exists at ten kilometres, so wind is an *exception* to decide and the other two are out; the daily 4 km and 5 km fields are 10–30 % valid and 0.2–1.5 TB each even compressed, so the gridded tier gains a sharded, tiled, compressed layout for grids finer than 0.25°; and several sources have ended or changed recently (Pathfinder SST stopped at 2023, IMERG V07 Final at 2025-09, NSIDC's AMSR2 line at 2025-08), so the family is built from the successors. The note gives the ledger, the storage design, the per-store row and byte arithmetic, a three-phase build ladder (phase A $\approx$ 0.5 TB, eight stores, keyless or hosted-runner credentials only), the falsifier that decides whether each store earned its place, and the decisions left to Chris.

1 Scope: the rule, and what it excludes

The rule. A source qualifies for family 1.gf if it is a *global* (or global-ocean) **observation** whose spatial support is ≤ 10 km — a point measurement counts — and whose temporal support is ≤ 5 days (a pentad or finer). A derived product is admitted only where it carries raw observations that are otherwise inaccessible (a merge of sensors whose individual level-2 files are impractical, a bias-corrected collation whose inputs are the observations). Monthly or coarser products enter only where something the goals need would otherwise be missing, and each such exception is named as one in §6.

Why this rule, in the programme's own terms. Family 7.2's grids are 0.25° and 1°, five-daily or monthly; family 10.1 added the point observations but no fine *fields*. E-078 measured why the grid is the wrong contract for fine data — a coarse value copied into fine cells is one measurement presented as many, a fine value averaged into a coarse cell throws away what the finer source was bought for — and E-077's ocean colour is the deliberate exception, box-averaged to 0.25° to get a channel onto the grid quickly. Family 1.gf is the fine half of that ledger, stored at its own resolution so the token's footprint fields can say what each value averaged.

What is already held and therefore not repeated. OISST and its sea-ice fraction at 0.25° (family 7.2, g025); NCEP at 1° (g100); GLORYS-derived interior fields and the Argo monthly climatology (rg100);

OC-CCI box-averaged to 0.25° (oc025); Argo core profiles (family 8); drifters, tropical moorings, SOCAT and all-mission along-track sea level (family 10.1); and the Global Fishing Watch daily 0.1° per-vessel fishing effort (family 10.2, E-081 — 11 km, so it sits just outside this family’s rule and stays where it is). Two of those are the coarse form of a store below — oc025 of oc4k, g025’s sea ice of seaice_asi — and both stay: the fine store is a new group, not a replacement, so a consumer that reads only tier G is unchanged.

Excluded by the rule, after checking the finest product that exists.

- **Sea-surface salinity.** Every product is ≥ 25 km on the grid and ≥ 40 km in effective resolution (RSS SMAP v6 “40 km”, ESA CCI SSS v5.5 at 25 km grid / 50 km stated, SMOS 25 km grid / 40–50 km footprint). Out.
- **Passive-microwave SST.** AMSR2 is a 0.25° grid from a ≈ 50 km footprint. Out; the infrared L3 products below are the fine-scale SST.
- **Ocean wind from scatterometers.** The finest gridded product is 0.125° (≈ 13.9 km, Copernicus WIND_GLO_PHY_L3_MY_012.005); the finest swath product is 12.5 km (EUMETSAT OSI SAF ASCAT L2, CC BY 4.0). Neither passes. Wind is the one forcing the ocean stores cannot do without, so it is carried as **exception E1** in §6 rather than dropped silently.
- **Gap-filled analyses:** OSTIA and MUR SST, DUACS gridded sea level, GLORYS, SMAP L4 soil moisture, the IMERG family’s model-like blending. They are derived; their inputs are what the stores below hold. IMERG is the one carried as an exception (E2), because the raw GPM swaths are 40 TB and cover the globe only every few days.

2 What fine granularity buys each goal

The four goals judge every addition (Chris, 2026-09-13). The mechanism per goal, so that the falsifier in §7 has something concrete to test:

goal	what happens at ≤ 10 km and ≤ 5 d that a 0.25° pentad cannot see	the stores that see it
El Niño	westerly wind bursts (days, hundreds of km), the convection that drives them (cloud tops cooling by tens of kelvin in hours), oceanic Kelvin waves as a sea-level ridge crossing the Pacific in two months, tropical instability waves at 1,000 km wavelength with 10 km fronts	irtb (cloud-top temperature every 30 min), swot and the held slatrack, sst_cci05, icoads (the pre-1979 record of Pacific SST and pressure), oc4k (equatorial upwelling shows in chlorophyll first)
carbon sink	blooms at 4 km over days; the biological pump’s timing; surface $f\text{CO}_2$ varies at 10 km scales along a ship track (held in socat); the interior’s stored carbon and the atmospheric gradient the flux works against	oc4k, bgcargoc (oxygen, nitrate, pH at depth), glodap (interior DIC, alkalinity, CFC ages), xco2 (the atmosphere’s side of the flux)
AMOC	deep convection in the Labrador and Irminger seas is set by the ice edge and by heat loss over days; boundary currents are 50–100 km wide with 10 km fronts; the density field that geostrophy reads comes from profiles, most of them not Argo before 2005	seaice_asi (6 km ice edge), swot (2 km sea level across the Gulf Stream and the boundary currents), wod (the pre-Argo hydrography, XBT lines), oceansites and deep_arrays (moored velocity and density time series), sst_cci05
SST	fronts are 2–10 km wide; diurnal warming is an hourly signal that the foundation-temperature analyses remove by design; coastal upwelling is a 10 km strip; the analyses the model has seen so far were <i>fitted</i> to these measurements, and a model that has only seen the fit cannot learn where the fit fails	sst_cci05 and sst_acspo02 (the level-3 measurements), icoads (ship and buoy SST since 1662, the record OISST and ERSST are fitted to), swh (wave-driven mixing)

Table 1: The mechanism each goal has at fine scale, and the stores that measure it.

3 The ledger

Every row was checked against the producer on 2026-09-16 (product pages, the NASA Common Metadata Repository, live directory and S3 listings, the Copernicus Marine STAC). “Raw” means level-2 or level-3 measurements binned or collated without gap-filling; “derived” means an analysis. Volumes marked *native* are the producer’s files; *stored* is this note’s estimate in the layout of §4, and §5 gives the arithmetic. A dagger (†) marks a number that could not be verified against the source and is an estimate.

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
oc4k	ESA OC-CCI v6.0 merged L3S, 4 km geographic 4320 × 8640: <code>chlora</code> , <code>kd490</code> , Rrs (6 bands), per-sensor <code>nobs</code> (PML)	1/24° (4.6 km)	daily; also 5-day, 8-day	1997-09-04 → 2026-06-30, quarterly updates	raw L3S	PML THREDDS / FTP, CEDA; no account	daily <code>chlora</code> 84–99 MB → ≈ 1 TB; all products 1.7 GB/day	ESA CCI free and open, cite
sst_cci05	ESA SST CCI v3.0 L3C per sensor, day and night, AVHRR NOAA-06.19 + MetOp-A/B and SLSTR-A/B; 0.05° 3600 × 7200 (Reading / Met Office)	0.05° (5.6 km)	daily, 2 passes	AVHRR 1979-07 → 2024; SLSTR 2016-06 →	raw L3C	CEDA HTTPS / OPeNDAP; no account (CDS mirror needs one)	AVHRR L3C 4 TB in 72.4 k files; SLSTR 272 GB	CC BY 4.0
sst_acspo02	NOAA ACSP0 L3S-LEO daily v2.81, super-collated VIIRS + AVHRR + MODIS, 0.02° 9000 × 18000 (NOAA STAR)	0.02° (2.2 km)	daily; AM and PM variants	2000-02-24 → present –2 months	raw L3S	PO.DAAC (Earthdata login); CoastWatch ERDDAP without	408 MB/day → 3.9 TB	NOAA open
seaice_asi	U. Bremen ASI v5.4 sea-ice concentration from AMSR-E / AMSR2 89 GHz (AMSR3 preliminary from 2026-06); polar-stereographic 6.25 km (1792 × 1216 N, 1328 × 1264 S) and 3.125 km	6.25 or 3.125 km	daily	2002-06-01 → 2011-10-04; 2012-07 → 2026-09-15	raw L3	HTTPS; no account	6.25 km ≈ 1.5 MB/day; AMSR2 record 8 GB (29 GB at 3.125)	“free for scientific use”, cite; redistribution terms to confirm
swot	SWOT L2 LR SSH Expert (version D), KaRIn 2 km grid on two 50 km swaths; 584 passes (292 orbits) per 20.86-day cycle (JPL / CNES)	2 km	per pass	2023-07-26 → (cycle 056 on 2026-09-13)	raw L2	PO.DAAC, Earthdata login, S3 us-west-2	32 MB/pass, NASA open 18.8 GB/cycle → ≈ 1 TB	open
swh	ESA CCI Sea State v4 L3 along-track significant wave height, track all missions Topex → Sentinel-6A, 1 Hz, as Copernicus <code>WAVE.GLO.PHY.SWH.L3.MY.014.005</code> (one daily multi-mission file)	≈ 7 km along track	1 Hz	1991-08-03 → 2023-12-31	raw L3	Copernicus Marine (account, as <code>slatrack</code>)	12–16 MB/day → 150–190 GB	Copernicus Marine licence
irtb	NCEP/CPC globally merged geostationary infrared brightness temperature, GES DISC <code>GPM.MERGIR</code> v1: 4 km, 9896 × 3298, 60°S–60°N, int16	4 km	30 min (two per hourly file)	1998-01 → now (–24 h); 251,616 files, none missing	raw merge	GES DISC HTTPS / OPeNDAP / S3, Earthdata login	≈ 16 MB per half hour → 6.96 TB	NASA open
icoads	ICoads Release 3.0 (1662–2014) + R3.0.3 near-real-time (2015 →), IMMA1: SST, air temperature, pressure, wind, humidity, cloud, waves (NOAA NCEI / NCAR)	point	per report	1662 → 2026-08	raw	NCEI HTTPS, no account (NCAR needs one)	45 GB + 117 GB gzip; ≈ 455 M reports to 2014† + ≈ 10 ⁸ /yr	CC BY 4.0 (GDEX) / NOAA open

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
wod	World Ocean Database 2023 + quarterly updates (Feb 2026): 18.63 M casts — OSD 3.26 M, CTD 1.13 M, MBT 2.43 M, XBT 2.36 M, MRB 1.28 M, DRB 0.27 M, PFL 2.75 M (Argo, excluded here), UOR 0.13 M, APB 2.06 M, GLD 2.97 M; T, S, O ₂ , nutrients, pH, alkalinity, CFCs, chlorophyll (NOAA NCEI)	point profile	per cast	1772 → 2026	raw	NCEI ragged netCDF per instrument-year, no account	167.5 GB; 121 GB without PFL	CC0
glodap	GLODAPv3 (July 2026) bottle data: 1,498,120 samples, 1,181 cruises; DIC 541 k, alkalinity 501 k, pH 332 k, O ₂ 1.30 M, nitrate 1.16 M, CFC-12 411 k (NCEI OCADS 0315582)	point bottle	per cruise	1972–2023	raw, adjusted	NCEI HTTPS, no account	1.0 GB CSV	CC BY 4.0
bgcargo	BGC-Argo synthetic profiles (Sprof): 2,963 floats, 407,999 profiles; DOXY 392 k, CHLA 195 k, BBP700 192 k, NITRATE 123 k, pH 102 k (Argo GDAC)	point profile	≈ 10 d	2002 →	raw	GDAC HTTPS/FTP, no account	≈ 15 GB	CC BY 4.0
oceansites	OceanSITES GDAC moored time series, the 52 non-TAO sites (TAO's 67 are held in gtmba); salinity at 103, currents at 44, O ₂ at 16 sites; hourly or better (NDBC / Ifremer)	point mooring	≤ 1 h	1980s → (six sites reporting in 2025)	raw	NDBC / Ifremer HTTPS, no account	26 GB without the 162 GB ALOHA cabled observatory	OceanSITES open policy
xco2	OCO-2 L2 Lite XCO2 v11.2r (+ v11.3r from 2024) and OCO-3 v11r: < 3 km ² soundings, 70–120 k good per day (GES DISC)	< 3 km ²	per sounding	2014-06 →; 2019-08 →	retrieval	GES DISC, Earthdata login	231 + 53 GB; 117 GB	NASA open
deep_arrays	instrument-level data of RAPID/MOCHA 26.5°N (BODC), OSNAP (Duke), SAMBA 34.5°S (AOML), MOVE 16°N (OceanSITES): T, S, velocity, bottom pressure at instrument depth	point mooring	15 min–1 h [†]	2004 / 2014 / 2009 / 2000 →	raw	public except RAPID's BODC terms, unverified	< 10 GB each [†]	open (RAPID to confirm)

Table 2: The candidates that pass the rule, as checked on 2026-09-16.

Checked and set aside. AVHRR Pathfinder v5.3 (4 km, 1981-08 → 2023-12-31): NCEI marks it “no longer updated”; the SST CCI L3C record covers the same sensors from 1979 with a licence, so Pathfinder is not needed. NSIDC AU_SI6: holds only 89 GHz brightness temperatures, no concentration, and NASA stopped updating it on 2025-09-01; AU_SI12 is 12.5 km. OSI SAF OSI-408-a: a 10 km grid from an 18 km footprint, 2016 →, CC BY 4.0 — the fallback for sea ice if Bremen’s terms refuse redistribution. NASA OB.DAAC single-sensor 4 km chlorophyll (MODIS-Aqua, VIIRS, PACE OCI from 2024-03): each is inside OC-CCI’s merge except PACE, which OC-CCI v6 does not yet include — a future **oc4k** channel, not a store. GDP hourly drifters v2.01.1 (197 M rows, 1987 → 2022-10): four times the rows of the held 6-hourly store, ends four years earlier, adds a diurnal SST decomposition; not now. Copernicus CORA (T/S profiles, 1950 → 2025-06, ≈ 90 GB): cleaner delayed-mode QC than WOD but T/S only and under the Copernicus licence; WOD is the pick for breadth and CC0, CORA the alternative if WOD’s QC costs too much in the probe. GPM DPR/GMI level-2 swaths (16.5 + 22.6 TB): the raw precipitation, but the globe is covered only every few days at 245 km swath width, so IMERG is carried instead (E2). TROPOMI

(NO₂, CO, CH₄ at 3.5×5.5 km, 23 + 8.5 + 2.5 TB): passes the rule and serves none of the four goals; a fire and fossil tracer for later. GOSAT via ACOS (10.5 km footprint): just over the line, 6.9 GB; would extend xco2 back to 2009-04 — an exception worth taking when xco2 is built (E3).

4 Storage: what family 10 already gives, and the one layout it lacks

Family 1.gf reuses family 10.1’s contract unchanged where it fits and adds one layout where it does not.

The registry. `tensors/family1_gf/family1gf.json` on `chfrank/earth-tensors`, schema version 2, the same top-level keys as `family10.json` (epoch 1982-01-01, `pentad_days` 5, the `bin_rule`, the `token_schema` with `log2_fp` and `log2_dt`, per-group `tier`, `cadence`, `C`, `channels`, `footprint`, `layout`, `prefix`, files with `sha256`, per-year counts, QC policy, `degraded`). The tier letters stay G, P and T so that a consumer’s dispatch is unchanged; a fine grid is tier G with `layout`: “sharded” (§4’s layout) rather than “binmajor”, and the reader named for tier G handles both by reading that field. Five per-group fields are new and all are load-bearing: `licence` with a `redistribution` value of `yes`, `attribution` or `no`; `frames_per_bin` and `frame_seconds` for gridded groups; `dtype` per group (float16 for measured fields, uint8 for classes and for brightness temperature stored as K–160); and `verdict`, empty until §7’s falsifier has run. A sharded group also names its `tile_grid.json` and `shard_index.npy`. Family 10.2’s registry — the current one, which inherits 10.1’s four stores by reference and adds `fishing` (E-081) — gains a one-line `siblings` list naming this file, family 1.0.tf’s and family 0.9.tf’s, so a consumer that already dispatches on `tier` discovers both families from the one entry point it knows. Nothing in family 10.1 or 10.2 is rewritten.

Tier P, with one widening. Every point store below ships the nine arrays of family 10.1 at schema 2 — `bin` int16, `time_s` int32 seconds, `lat`, `lon` float32, `values` float16 $[N, C]$ in raw units, `platform` int64, `qc` uint8, `fp` float16 $[N, 2]$, `bin_offsets` int64 — sorted by (`bin`, `time_s`), negative bins before 1982, longitude in $[-180, 180)$ enforced in float32, one row costing $27 + 2C$ bytes. Profiles put the vertical axis into C exactly as family 8 does for Argo: the 16 Roemmich–Gilson pressures per variable (the standard levels of the Roemmich–Gilson Argo climatology, 2.5 to 2,000 dbar), NaN where the cast did not reach. The one widening: an int32 second since 1982 reaches back only to 1913-12-13, and `icoads` begins in 1662 and `wod` in 1772. A store whose record starts before 1914 declares **schema 3**, identical except that `time_s` is int64; a row then costs $31 + 2C$ bytes, `bin` (int16, ± 449 years) is unchanged, and `ml/family10_store.py` gains one dtype branch. Stores inside the int32 range stay at schema 2.

Tier G-fine, the new layout. Family 7.2’s tier G is one bin-major `.npy` per group, read by one HTTP range of one bin: 14.5 MB at 0.25° for seven channels. At 0.05° one daily frame of one channel is 51.8 MB and a day of `sst_acspo02` is 324 MB per channel; a cone anchored at one point needs a few hundred kilometres of it, and the daily fields are 10–30 % valid, the rest cloud, night or land. A flat uncompressed array would cost 5–10 times the bytes it carries and force every anchor to read a whole globe. So groups on a grid finer than 0.25° use a **sharded, tiled, compressed** layout:

- One **shard** per (group, bin): `<prefix>/<yyyy>/<bin>.shard`. Inside it, F **frames** per bin (`frames_per_bin` = 1 for a pentad field, 5 for daily, 10 for day/night, 40 for three-hourly, 240 for half-hourly), each frame cut into **tiles** of 256×256 pixels of the group’s native grid (12.8° at 0.05° ; 10.7° at $1/24^\circ$; 1,600 km on the 6.25 km polar grid), each tile all C channels as float16 in raw units, NaN for unobserved, zstd-compressed.
- One **index** per group: `shard_index.npy`, int64 $[n_{\text{bins}}, F, n_{\text{tiles}}, 2]$ of (byte offset, byte length) inside the bin’s shard, plus `tile_grid.json` (rows, columns, the lat/lon of every tile’s corner). An empty tile (all NaN — open ocean for a land product, the tropics for sea ice) has length 0 and is never stored. Tiles are float16 unless the group’s `dtype` says otherwise.
- A **read** is one HTTP range per (bin, frame, tile) the cone touches: 20–130 KB compressed for a tile with 20 % valid pixels, measured by the probe (§7) before anything is built. The bin’s whole frame is `n_tiles` ranges; a cone of 1,400 km reach touches four to nine tiles at 0.05° and up to about 36 at 0.02° .

- The Hub’s limits shape this: under 100k files per repository and 10k per folder, so shards are per bin (about 3,450 for a group reaching back to 1979) in year folders, never per tile; files under 200 GB, which no shard approaches.

The frame carries its own timestamp (`frame_seconds` in the registry: 86,400 for daily, 10,800 for three-hourly; for day/night passes the local overpass time is a channel), and the token’s Δt is measured from the end of the anchor’s bin to the frame’s time, as tier P does. A gridded value is one token at the pixel centre with $\log_2 \text{fp} = \log_2(\text{pixel km}/27.83)$ and $\log_2 \text{dt} = \log_2(\text{frame days}/5)$; nothing is copied into neighbouring pixels. On the floor: family 10.1 writes $\log_2 f_p = -4$ for a point instrument, and this note keeps that as the *label* for “no areal support”; a pixel with a real support carries its true value even below -4 (a 1 km pixel is -4.8 , a 300 m pixel -6.5), and likewise a real temporal support (an hour, -6.9) is written as it is while an instantaneous sample keeps the -4 label. E-078 §2 quoted OC-CCI’s 4 km as -2.8 ; the grid is $1/24^\circ = 4.64$ km, so -2.58 is the value used here.

Which tier a fine field belongs in is geometry first, bytes second. A row of tier P costs $27 + 2C$ bytes; a stored pixel of a compressed G-fine tile costs about $2C \times 1.2$ bytes and an unobserved pixel almost nothing, so on bytes alone a compressed grid beats rows at any valid fraction (an *uncompressed* grid would lose below $v = 2C/(27 + 2C)$, 7% at $C = 1$). What decides is whether the source *has* a fixed grid: SWOT’s 2 km cells lie along two swaths whose position changes every pass, so there is no raster to tile and `swot` is a P store; a 4 km chlorophyll field at 10–15% valid is on one grid every day and is G-fine, and a cone over the tropics then reads a few tiles instead of millions of rows. The probe measures v per source; the registry records the tier and the layout.

Tier T is not used by this family. Nothing global at ≤ 10 km is imagery in the tile sense — the 300 m ocean-colour and Sentinel swaths are coastal or terrestrial and belong to family 1.0.tf (and, as a published embedding in place of the raw imagery, family 0.9.tf). This departs from E-078 §3, which placed 300 m–1 km radiometry in tier T: a *gridded level-3* product at those scales is a raster on one fixed grid and fits the sharded tier-G layout, whereas tier T is for swath imagery whose footprint moves every pass. The one candidate that approaches tier-T volume, `irtb` at 30 minutes, is handled by scope (§5) rather than by a token cache.

Two-machine builds, unchanged. Credentials — Earthdata login, Copernicus Marine — live in repository secrets and reach only hosted runners; rented boxes are keyless and assemble from Hub parts, exactly the `slatrack` pattern (six hosted lanes fetch and push per-year parts with `done.json` written last after a restore-hash; a keyless verified box pulls, assembles by streaming, publishes with restore-verification). Every fetch lane refuses a listing that came back empty or a download that came back short and records any relaxation in `degraded`; there is no “download failed, skip” path, per Chris’s 2026-09-14 rule.

5 The stores in numbers

Row counts and stored bytes below are this note’s estimates from the verified native facts; every one is a hypothesis the one-month probe of §7 replaces with a measurement before a build is dispatched. Bytes per P row are $27 + 2C$ plus headers; G-fine bytes assume `zstd` at the stated valid fraction (a NaN tile compresses to nothing, a valid pixel to $\approx 2C$ bytes plus 10–20% overhead), so **stored** $\approx$ **pixels** $\times$ **frames** $\times v \times 2C \times 1.2$.

Three arithmetic notes. `swot`’s 16 billion rows are eight times `slatrack`’s, because a swath is a grid and a nadir track is a line; at 33 bytes a row the store would be about 0.5 TB. The alternatives — keep the 250 m unsmoothed product (25 TB native) or subsample the 2 km grid — are respectively impossible and against the rule, so `swot` is built as a P store in phase B, once the Hub conversation about terabyte repositories has happened. `irtb` is the largest raw source in the ledger and its scientific value for El Niño is in the tropics, so phase B takes 30°S–30°N three-hourly (0.7 TB) and the full half-hourly 60°S–60°N record is a phase-C decision; the frame count, not the pixel count, is what the design has to hold. `sst_cci05` is built by the family’s own builder from the per-sensor L3C files, choosing the highest quality level per pixel per pass — a super-collation, which is what NOAA’s ACSPO L3S does at 0.02° from 2000; the CCI route is taken first because it reaches 1979 and is CC BY 4.0, and `sst_acspo02` follows for the 2 km detail.

store	tier	channels (C)	footprint ($\log_2 f_p, \log_2 f_t$)	rows or frames	stored	goals	phase
seaice_asl	G-fine	1: sic (%) on two polar grids as two sub-groups N/S	(-2.15, -2.32)	3.86 M px $\times$ 8,700 d; $v \approx 0.4$ of the polar grids	≈ 30 GB	AMOC, SST	A
oc4k	G-fine	3: log_chl, kd.490, nob; +6 Rrs bands in phase B	(-2.58, -2.32)	37.3 M px $\times$ 10,500 d; $v \approx 0.10$ -0.15	0.3-0.4 TB ($C = 3$); ≈ 1.1 TB with Rrs	carbon, SST, El Niño	A
icoads	P	8: sst, airt, slp, wind.u, wind.v, dewpt, wave.h, cloud	(-4, -4)	≈ 1.5 B [†] (455 M to 2014 + NRT)	≈ 75 GB	SST, El Niño, AMOC	A
wod	P	48: T, S, O ₂ at the 16 RG pressures; platform = cast id, instrument in qc bits	(-4, -4)	15.9 M casts (WOD without PFL)	2 GB	AMOC, SST	A
glodap	P	9: depth, DIC, TA, pH, O ₂ , NO ₃ , PO ₄ , SiO ₄ , CFC-12	(-4, -4)	1.50 M bottles	80 MB	carbon, AMOC	A
bgcargo	P	80: DOXY, NITRATE, PH, CHLA, BBP700 at 16 pressures	(-4, -4)	408 k profiles	80 MB	carbon	A
oceansites	P	5: depth, T, S, u, v; one row per (site, hour, instrument)	(-4, -6.9)	$\approx 10^8$ [†]	≈ 4 GB	AMOC, El Niño, carbon	A
xco2	P	3: xco2, xco2_unc, surface (land/water)	(-4, -4)	≈ 400 M good soundings (OCO-2) + OCO-3	≈ 18 GB	carbon	A
swot	P	3: ssh.karin, sig0, quality	(-3.80, -4)	≈ 0.5 M px/pass $\times$ 584/cycle $\times$ 55 cycles ≈ 16 B	≈ 0.6 TB	AMOC, El Niño	B
swh	P	3: swh, swh.denoised, swh_unc	(-2.0, -4)	≈ 2 B (the slatrac missions, 1991-2023)	≈ 70 GB	SST (mixing), read-out	B
sst_cci05	G-fine	3: sst, quality, dtime; day and night as 2 frames/day, the builder super-collating sensors by best quality	(-2.32, -2.32)	25.9 M px $\times$ 2 $\times$ 16,200 d; $v \approx 0.2$	≈ 1.2 TB	SST, all	B
irtb	G-fine	1: tb (uint8, K-160), 30°S-30°N, 3-hourly ($F = 40$)	(-2.80, -7.9)	16.3 M px $\times$ 8/day $\times$ 10,500 d; $v = 1$, but Tb compresses $\approx 2 \times$	≈ 0.7 TB	El Niño, SST	B
precip01 (E2)	G-fine	2: precip, quality at 0.1°, daily	(-1.32, -2.32)	6.48 M px $\times$ 10,100 d; $v = 1$, but zeros compress 10^7 - 10^8 [†]	0.2-0.3 TB	El Niño, AMOC	B
deep_arrays	P	6: depth, T, S, u, v, bpr	(-4, -6.9)	10^7 - 10^8 [†]	< 5 GB	AMOC	B
sst_acspo02	G-fine	2: sst, quality at 0.02°	(-3.64, -2.32)	162 M px $\times$ 9,700 d; $v \approx 0.3$	≈ 2.3 TB	SST	C
wind12 (E1)	P	3: u10, v10, quality at 12.5 km swath cells	(-1.15, -4)	≈ 4 M cells/day/satellite $\times$ 27 yr ≈ 50 B	≈ 1.7 TB (or ≈ 50 GB as pentad 0.125° frames)	all	C
irtb full	G-fine	60°S-60°N, half-hourly ($F = 240$)		32.6 M px $\times$ 48 $\times$ 10,500 d	≈ 8 TB		C

Table 3: Estimated size of every store in the layout of §4, and the phase it is proposed for. Phase A totals ≈ 0.5 TB; phase B ≈ 3.5 TB more; phase C is where the Hub’s “contact us for terabytes” threshold is certainly crossed.

6 The exceptions, named

- **E1 — scatterometer wind at 12.5 km.** Fails the rule by 25% and is the only observation of the wind that forces every ocean store here. Proposed: admit in phase C as `wind12`, a P store of OSI SAF ASCAT L2 12.5km swath cells (CC BY 4.0; Metop-A 2007–2014 CDR, Metop-B 2012 →, Metop-C 2019 →) with QuikSCAT (1999–2009) from the Copernicus 0.25° line for the earlier years. The pick between the P form (1.7 TB, every cell) and pentad 0.125° frames (90 GB, a mean) is Chris’s; this note’s pick is the P form restricted to 2007 → first.
- **E2 — IMERG at 0.1° (≈ 11 km), daily.** A derived merge, gauge-calibrated in the Final run; admitted because the raw swaths cannot cover the globe daily and precipitation is the freshwater flux the AMOC goal needs. The V07 Final record ends 2025-09-30; V08 Final is expected in the second half of 2026, so the store is built when V08 lands or from V07 Final + V07 Late for the tail, with the run recorded per row in `qc`.
- **E3 — GOSAT via ACOS at 10.5 km.** Over the line by half a kilometre, 6.9 GB, and it extends the column-CO₂ record from 2014 back to 2009-04. Admitted into `xco2` with `platform` distinguishing the instrument and `fp` carrying its own footprint.
- **Monthly:** nothing. Every goal’s monthly quantities (RAPID transport, the Niño indices, SeaFlux) are targets or are already held coarse; no monthly source is critical to this family.

7 The build ladder, the probe, and the falsifier

Phase A — eight stores, ≈ 0.5 TB, keyless or hosted-runner credentials only. `seaice_asi`, `oc4k`, `icoads`, `wod`, `glodap`, `bgcargo`, `oceansites`, `xco2`, plus the registry and the `siblings` line in family 10.2’s. Seven of the eight fetch without any account; `xco2` needs an Earthdata login on a hosted runner; `oc4k` is the one G-fine build large enough to need a box for assembly. Expected cost, from the 10.1 precedent: the P stores \$0 on hosted runners; `oc4k` one verified box for a few hours, \$2–5; the whole phase under \$10 if nothing is re-fetched.

Phase B — the terabyte stores. `swot`, `sst_cci05`, `irtb` (tropics, three-hourly), `precip01`, `swh`, `deep_arrays`, and the six Rrs channels of `oc4k`. ≈ 3.5 TB; \$20–40 in box time at 10.1’s measured rates, dominated by upload; the Hub should be asked about a terabyte-scale public repository before this phase (the docs say “contact us for large repos”).

Phase C — decisions, not builds, until A and B have verdicts. `sst_acspo02`, `wind12`, the full `irtb`, PACE into `oc4k`.

The probe, before any build. For each source, one month (January 2015, as E-079 used) fetched on a hosted runner through the real adapter, measuring: rows per day or valid fraction per frame; compressed bytes per tile at the chosen tile size; listing completeness against the producer’s own count; fetch throughput; and for G-fine, the byte length of a cone-sized read. Family 10 learned the cost of building on the spec’s numbers — a time column 84 times coarser than the sampling, 908 rows in the wrong pentad — so **no store is dispatched on a size that has not been measured**, and the probe’s numbers replace this note’s in the registry.

The falsifier per store. A store earns its place if, at anchors where it has coverage, adding its tokens to the cone lowers the held-out masked-token loss on the goal’s read-out (§2’s mapping: the Niño-3.4 target for El Niño, the `socat`-scored $f\text{CO}_2$ and `oc025` colour targets for carbon, the RAPID transport and corridor targets for AMOC, the OISST target for SST) against the same anchors, same seed, without it — three seeds, because a new tier buys its own replication (`m1/CLAUDE.md` §3b). Parity means the coarse tensor already carried what the fine store adds at pentad cadence, and the store is then justified only as a fine *target* (E-078 §5), which is a different and cheaper claim. The verdicts are written per store into the registry’s `verdict` field so a consumer can see which fine stores the programme found to matter.

Verification, as for 10.1. Every array’s sha256 against `store.json`; N from file sizes; `bin` monotone and equal to `floor_divide(time_s, 432000)`; the CSR index reproduced; and, new for G-fine, every shard’s index entries decompressing to tiles of the declared shape with the declared NaN fraction, and per-year frame counts equal to the producer’s listing (a day absent upstream is a zero-length frame with a counter, never a silent gap).

8 Decisions for Chris

Laid out with this note’s pick named, all reversible before phase A is dispatched.

1. **The name.** “Family 1.gf” as given; if “1” was meant as “10” (family 10.gf, the fine sibling of 10.1), the change is the prefix and the registry name, nothing else. Pick: as given.
2. **G-fine layout** — sharded tiled zstd shards with an index, versus a flat `.npy` per bin. Pick: sharded; the flat form costs 5–10× the bytes and makes every anchor read a globe.
3. **Daily frames or pentad means** for the G-fine stores. Pick: daily frames ($F = 5$); the rule asks for ≤ 5 days and the pentad mean is a derived quantity the consumer can form.
4. **Sea ice from Bremen ASI** (6.25 km, 2002 →, terms to be confirmed by email before publishing) or OSI SAF OSI-408-a (10 km grid, CC BY 4.0, 2016 →). Pick: Bremen, with OSI SAF as the fallback if the answer is no.
5. **SST**: CCI L3C super-collated by our builder at 0.05° first, ACSPO L3S 0.02° later. Pick: as stated.
6. **Exceptions E1–E3**: admit wind at 12.5 km (phase C), IMERG daily (phase B), GOSAT into `xco2`. Pick: admit all three; refuse SMAP-style “9 km grids from 36 km footprints” anywhere in this family.
7. **irtb scope**: tropics three-hourly first. Pick: yes.
8. **WOD (CC0) or CORA (Copernicus)** for the non-Argo profiles. Pick: WOD.
9. **The Hub**: write to Hugging Face about a terabyte-scale public dataset before phase B. Pick: yes, with phase A’s measured sizes in hand.

Terms used above, in one line each

AMOC the Atlantic Meridional Overturning Circulation, the Atlantic’s northward heat-carrying flow, read out as transport at 26.5°N (RAPID). **Tier G / P / T** family 10’s three storage tiers: gridded dense fields; points, profiles and tracks; tiles of imagery. **CSR** compressed sparse row — an offsets array saying where each five-day bin’s rows start. **zstd** Zstandard, a fast lossless compressor. **STAC** SpatioTemporal Asset Catalog, the metadata standard producers publish scene listings in. **L2 / L3 / L3C / L3S / L4** processing levels: retrievals on the instrument’s swath; binned onto a grid; collated per sensor; super-collated across sensors; gap-filled analysis (derived). **Rrs** remote-sensing reflectance, the water-leaving colour per wavelength. **Kd₄₉₀** the diffuse attenuation of light at 490 nm, a water-clarity measure. **OC-CCI** ESA’s Ocean Colour Climate Change Initiative merged record (PML, Plymouth). **CEDA / THREDDS / OPeNDAP** the UK atmospheric data archive, and two protocols for reading subsets of netCDF files over HTTP. **AVHRR / SLSTR / VIIRS / MODIS** infrared radiometers on NOAA and MetOp, Sentinel-3, JPSS and Terra/Aqua satellites. **ACSPO** NOAA’s SST processing system. **ASI** the ARTIST sea-ice algorithm using 89 GHz microwave channels (AMSR-E, AMSR2, AMSR3 radiometers). **KaRIn** SWOT’s Ka-band radar interferometer that measures sea level across two 50 km swaths. **GES DISC / PO.DAAC / NCEI / AOML / NDBC / BODC** NASA’s, NOAA’s and the UK’s data centres. **ICODAS / IMMA1** the ship-and-buoy marine report archive and its record format. **WOD** the World Ocean Database; its instrument codes: OSD bottle casts, CTD conductivity-temperature-depth profiles, MBT/XBT mechanical and expendable bathythermographs, MRB/DRB moored and drifting buoys, PFL profiling floats (Argo), UOR undulating recorders, APB tagged animals, GLD gliders. **GLODAP** the Global Ocean Data Analysis Project bottle-data synthesis; **DIC / TA** dissolved inorganic carbon and total alkalinity; **CFC** chlorofluorocarbon tracers that date water masses. **BGC-Argo / Sprof / DOXY / BBP700** biogeochemical Argo floats, their merged profile files, dissolved oxygen, and particle backscatter at 700 nm. **GDAC** the Argo global data-assembly centre. **OceanSITES / TAO** the open-ocean moored-time-series network and the tropical Pacific mooring array (held in family 10.1’s `gtmba`). **OCO-2/3 / GOSAT / ACOS** NASA’s and Japan’s column-CO₂ satellites and NASA’s retrieval of GOSAT. **TROPOMI** the trace-gas imager on Sentinel-5P. **OSI SAF / ASCAT** EUMETSAT’s ocean-and-ice facility and the MetOp scatterometer. **OSTIA / MUR / DUACS / GLORYS** gap-filled SST analyses (Met Office, JPL), the gridded altimetry analysis, and the ocean reanalysis — all derived. **DPR / GMI** GPM’s precipitation radar and microwave

imager. **CORA** Copernicus' quality-controlled T/S profile collection. **PACE OCI** NASA's 2024 hyperspectral ocean-colour instrument. **E-077 / E-079** the experiments that built ocean colour into family 7.1 and family 10's first four stores. **g025 / g100 / rg100 / oc025** family 7.2's four gridded groups: 0.25° ocean surface, 1° atmosphere and land, 1° monthly interior, 0.25° ocean colour.

References

- [1] blauewelt/earth (2026). Family 10.1: the granularity-aware tensor, as built. Design note, 16 September 2026, [ml/paper/notes/family10.pdf](#).
- [2] blauewelt/earth (2026). E-078: family 10 — one schema for every granularity. [ml/plans/E078_multi_granularity.md](#).
- [3] blauewelt/earth (2026). E-079: family 10, the first data pull. [ml/plans/E079_family10_point_stores.md](#).
- [4] Sathyendranath, S. et al. (2019). An ocean-colour time series for use in climate studies: the experience of the Ocean-Colour Climate Change Initiative. *Sensors* 19, 4285. Data: OC-CCI v6.0, PML, [oceancolour.org](#).
- [5] Embury, O. et al. (2024). Satellite-based time-series of sea-surface temperature since 1980 for climate applications. *Sci. Data* 11, 326. Data: ESA SST CCI / C3S v3.0, CEDA. CC BY 4.0.
- [6] Spreen, G., Kaleschke, L., Heygster, G. (2008). Sea ice remote sensing using AMSR-E 89-GHz channels. *J. Geophys. Res.* 113, C02S03. Data: University of Bremen, [data.seaice.uni-bremen.de](#).
- [7] NASA/CNES (2025). SWOT Level 2 KaRIn Low Rate Sea Surface Height, Expert, Version D. PO.DAAC.
- [8] Janowiak, J., Joyce, B., Xie, P. (2017). NCEP/CPC L3 half hourly 4 km global (60S–60N) merged IR V1. GES DISC, DOI 10.5067/P4HQB9N27EQU.
- [9] Freeman, E. et al. (2017). ICOADS Release 3.0: a major update to the historical marine climate record. *Int. J. Climatol.* 37, 2211–2232.
- [10] Boyer, T. P. et al. (2024). World Ocean Database 2023. NOAA Atlas NESDIS 97.
- [11] GLODAP (2026). Global Ocean Data Analysis Project version 3. NCEI accession 0315582, DOI 10.25921/m6tp-mj50. CC BY 4.0.
- [12] Crisp, D. et al. (2017). The on-orbit performance of the Orbiting Carbon Observatory-2 instrument. *Atmos. Meas. Tech.* 10, 59–81.